Exp No:12 Experiment to detect outliers in a given data set.

AIM:

To detect outliers in a given data set.

Description:

Understand the procedure to identify the outliers in a given dataset.

CODE:

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("student_marks_outliers.csv")
df
array = df["Marks"].values
array
array.mean()
print("25th Percentile:", np.percentile(array, 25))
print("50th Percentile:", np.percentile(array, 50))
print("75th Percentile:", np.percentile(array, 75))
print("100th Percentile:", np.percentile(array, 100))
Q1 = np.percentile(array, 25)
Q3 = np.percentile(array, 75)
IQR = Q3 - Q1
print("Q1 =", Q1)
print("Q3 =", Q3)
print("IQR =", IQR)
lower_limit = Q1 - (1.5 * IQR)
upper_limit = Q3 + (1.5 * IQR)
print("Lower Limit:", lower_limit)
```

```
print("Upper Limit:", upper_limit)
outliers = array[(array < lower_limit) | (array > upper_limit)]
print("Outliers:")
print(outliers)
def outDetection(array):
   Q1 = np.percentile(array, 25)
   Q3 = np.percentile(array, 75)
   IQR = Q3 - Q1
   lr = Q1 - (1.5 * IQR)
   ur = Q3 + (1.5 * IQR)
   return lr, ur
 lr, ur = outDetection(array)
print("Lower Range:", lr)
print("Upper Range:", ur)
sns.histplot(array, bins=8, kde=True)
plt.title("Distribution of Student Marks")
plt.xlabel("Marks")
plt.ylabel("Count")
plt.show()
```

output:

25th Percentile: 72.0

50th Percentile: 79.0

75th Percentile: 87.0

100th Percentile: 130.0

Q1 = 72.0

Q3 = 87.0

IQR = 15.0

Lower Limit: 49.5

Upper Limit: 109.5

Outliers:

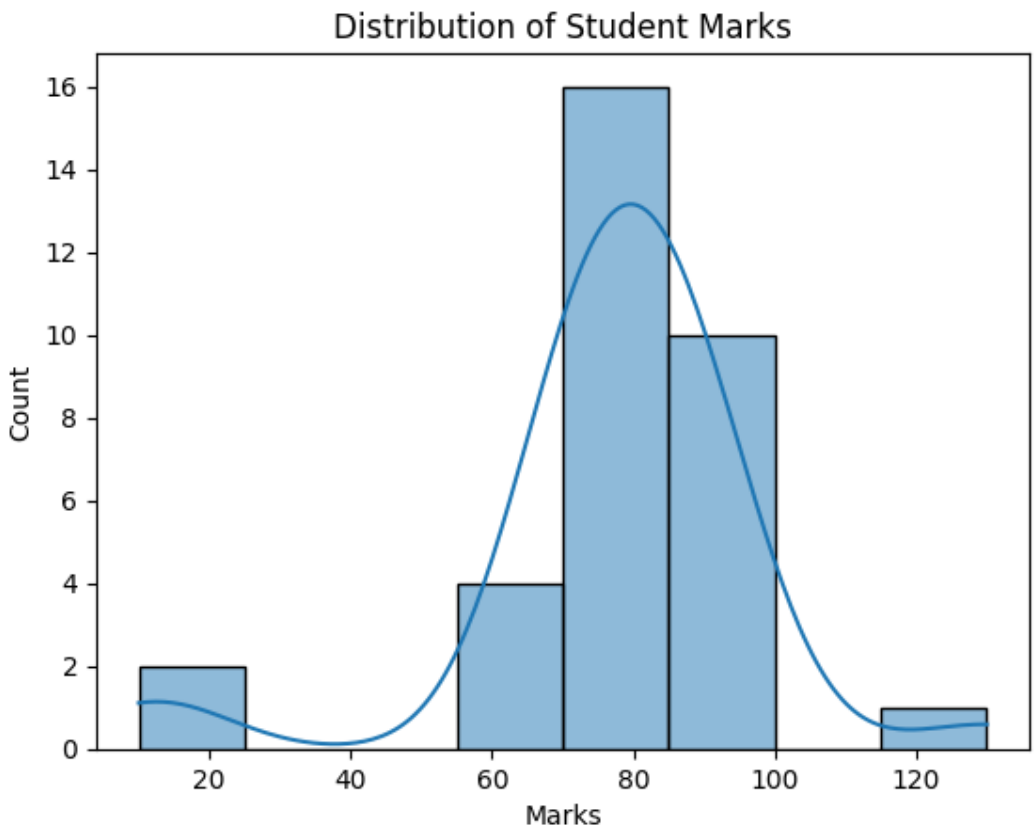
[10 15 130]

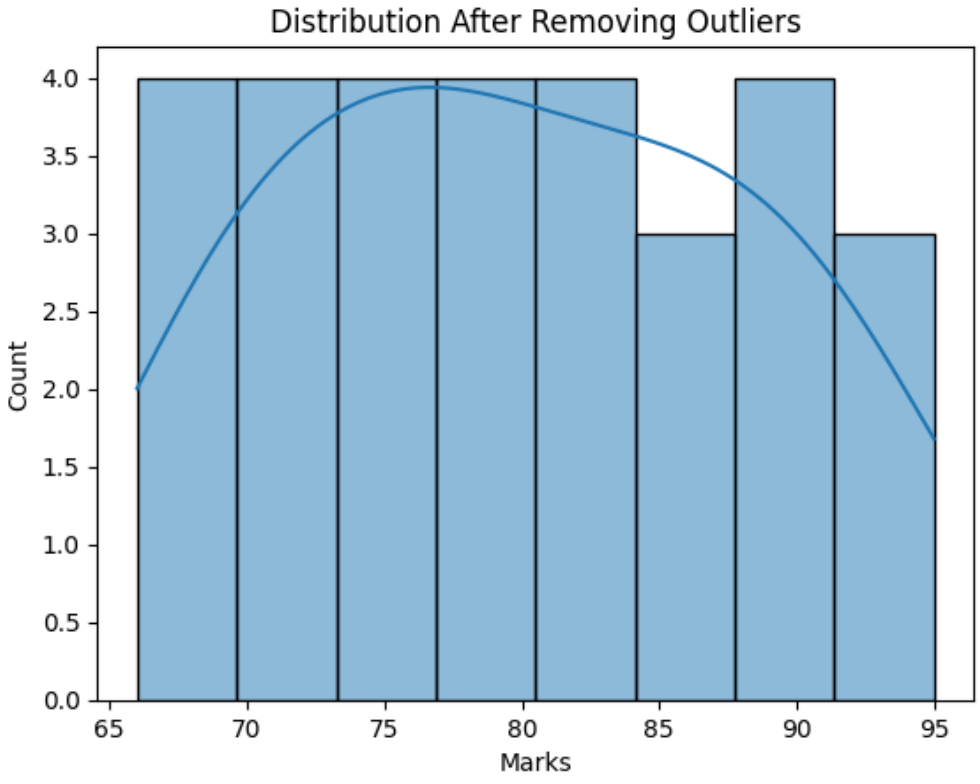
Lower Range: 49.5

Upper Range: 109.5

array([72, 68, 75, 81, 77, 69, 85, 74, 80, 78, 83, 71, 76, 88, 79, 82, 73,

86, 90, 84, 67, 70, 92, 87, 89, 94, 66, 91, 95, 76])


Distribution of Student Marks
Count
Marks
0
2
4
6
8
10
12
14
16
20
40
60
80
100
120


Distribution After Removing Outliers
Count
Marks
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
65
70
75
80
85
90
95

Result:

Outliers in the student marks dataset were successfully identified using the IQR method and the effect of removing them was visualized.